

AGENDA FOR THE DAY-4

LOGIC BUILDING

- ✗ Revise loops
- ✗ Ternary operator ? :
- ✗ Package – java.lang, java.util, java.io
- ✗ Taking input from Scanner
- ✗ Patterns programs

PACKAGE

- ✖ As per oracle official docs, a **package** is a namespace that organizes a set of related classes and interfaces.
- ✖ You can think of packages as similar classes to different folders on your computer
- ✖ package names are written in all lowercase, and to avoid naming conflicts with other organizations, companies typically use their reversed internet domain as a prefix — e.g., `com.example.mypackage`, `com.whatsapp.media`
- ✖ Types— built-in and user defined.

PACKAGE

java.lang

- ✗ This is the most fundamental package in Java.
- ✗ Oracle's docs describe it as providing classes that are integral to the design of the Java programming language.
- ✗ It's the *only* package automatically imported into every Java source file, without needing an explicit import statement.
- ✗ It contains **Object**, the root of every class hierarchy — meaning every single Java class you ever write has an implicit dependency on this package.

Major content –

- ✗ **Object** — the root class of the entire class hierarchy; every class in Java implicitly extends it.
- ✗ **System** — access to system resources like `System.out`, `System.exit()`, environment properties.
- ✗ Exception classes, `String`, Wrapper classes etc

PACKAGE

java.util

- ✖ Oracle describes this as the package containing the **collections framework**, legacy collection classes, event model classes and miscellaneous utility classes.

Collections Framework interfaces — List, Set, Map, Queue, Deque, Collection, Iterator, **Scanner** — commonly used for reading user input.

java.io

- ✖ Oracle's docs describe this package as providing classes for **system input and output through data streams, serialization, and the file system**.
InputStream/OutputStream, Buffered I/O

STRING

- ✖ The **String** class in Java is used to represent a sequence of characters. It is one of the most commonly used classes in Java and belongs to the **java.lang** package, which is imported automatically.
- ✖ Its Immutable. Means once a String object is created, its value cannot be changed.
- ✖ Any operation that seems to modify a string actually creates a new String object.

```
String str = "Hello";
```

```
str = str.concat(" World");
```

```
System.out.println(str); // Hello World
```

String Creation- using string literal– `String s1 = "Java";`

Using the new keyword:- `String s2 = new String("Java");`

STRING

- ✗ String is not a primitive data type.
- ✗ We can declare String as a data type

```
String name = "cdac";
```

```
String city = "mumbai";
```

```
int age = 20;      // int is primitive
```

```
char grade = 'A';  // char is primitive
```

```
boolean pass = true; // boolean is primitive
```

```
String name = "Rahul"; // String is a reference type
```

TAKING INPUT FROM USER

✗ Using Scanner class-

- ✗ Scanner class in Java is used mainly to take input from the user through the keyboard.
- ✗ It belongs to the java.util package.
- ✗ Before using Scanner, you need to import it: `import java.util.Scanner;`
- ✗ Once imported, then create Scanner object → `Scanner sc = new Scanner(System.in);`
- ✗ Use `nextInt()` method for int or as required.
 - `nextInt()` → int
 - `nextDouble()` → double
 - `nextFloat()` → float
 - `nextLong()` → long
 - `nextBoolean()` → boolean
 - `next()` → one word
 - `nextLine()` → complete line

TAKING INPUT FROM USER

```
import java.util.Scanner;
```

```
class ScanClass{
```

```
    public static void main(String args[]){
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter your name :");
```

```
        String name = sc.nextLine();
```

```
        System.out.println("Hello " + name + "Welcome to CDAC Mumbai");
```

```
    }
```

```
}
```